

Investigating Semester Difficulty of an Undergraduate Industrial & Systems Engineering Degree: A Case Study and Surveying Approach

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Abstract

This study examines the distribution of workload and course difficulty across the Industrial & Systems Engineering (ISE) curriculum at the University of Oklahoma using two complementary methods: a five-semester case study with 2,668 daily study-time observations across 24 courses, and a Qualtrics survey of 24 upper-division ISE students. An “Easiness Score” combining ending grade, average study session length, and study duration is constructed for each course, with study time treated as a random variable so that distributional inference is well-defined; Monte Carlo simulation ($n = 120$) is used to recover the survey-side distribution. Kruskal-Wallis tests reject the null of equal Easiness Scores across both courses ($p < 2.2 \times 10^{-16}$) and semesters ($p = 5.6 \times 10^{-5}$) in the case study, and 62 of 276 pairwise course comparisons remain significant after Bonferroni adjustment. Applied Engineering Statistics, Data-Driven Decision Making I, and Deterministic Systems Models emerge as the three most difficult courses across both methods, and Junior-year Fall is the most challenging semester. 71% of surveyed students report that semester difficulty is unevenly balanced, and 50% indicate that the course distribution has affected their anticipated graduation date. We recommend separating DDDM I and DSM into different semesters and shifting Applied Engineering Statistics back one term to balance workload while respecting the existing prerequisite chain.

1 Introduction

1.1 Background and Motivation

Burnout and delayed graduation are persistent problems among engineering undergraduates, and uneven course loads across semesters are frequently cited as a contributor. When individual semesters carry roughly comparable workloads, students experience less stress, achieve higher grades, and are more likely to graduate on time ([Huntington-Klein and Gill,](#)

2021; Karma et al., 2021). Within the Gallogly College of Engineering at the University of Oklahoma, the Industrial & Systems Engineering (ISE) curriculum concentrates several quantitatively demanding courses—Applied Engineering Statistics, Deterministic Systems Models, and Data-Driven Decision Making I—in the same one or two semesters. Whether this concentration is warranted by prerequisite structure or could be redistributed without breaking the flowchart is an empirical question that has not been answered for this program.

Prior work suggests that perceived workload does not align well with credit hours (Pardos et al., 2023), that lighter loads support better grades especially in early semesters (Szafran, 2001), and that excess time-to-degree itself depresses GPA and labor-market bargaining power (Brugiavini et al., 2020; Bound et al., 2012). Together these findings motivate examining course difficulty within ISE on a per-course basis using a measure that captures both effort and outcome rather than relying on credit hours alone.

1.2 Literature Review

Huntington-Klein and Gill (2021) used administrative data and within-student fixed-effects quantile regression at a four-year university to show that more evenly distributed workloads produce higher overall grades and lower stress than semesters concentrated with heavy loads. Karma et al. (2021) employed a sequential mixed-methods approach—an online survey of perceived academic workload followed by qualitative coding of student reflections—and reached a similar conclusion: inconsistent workloads predict burnout and reduced satisfaction. Aloia and McTigue (2019) ran a Qualtrics-based Likert survey at a southern university and found a direct negative association between course load, time pressure, and psychological well-being. Szafran (2001) sampled 487 students from a university registry and found that lighter loads correlated with higher final grades, particularly in early semesters.

A second strand questions whether credit hours measure workload accurately at all. Pardos et al. (2023) compared LMS-derived effort metrics with enrollment data and showed that students’ perceived workload diverges meaningfully from credit-hour-based institutional metrics. On the other side, Attewell and Monaghan (2016) followed 8,230 freshmen over five years and found that students who began with heavier loads (15+ credits) graduated sooner, and Phipps and Amaya (2023) used randomly assigned course loads to show a non-monotonic relationship: small increases above 15 credits raise GPA and shorten time-to-degree, but larger overloads disproportionately harm underprepared students.

Time-to-degree itself is a meaningful outcome. Bound et al. (2012) documented a 45-year decline in the share of students completing in four (or even five) years using semi-parametric weighting. Brugiavini et al. (2020) found that excess time-to-degree is associated with lower GPA and weaker labor-market outcomes. Salas-Velasco (2024) applied

Data Envelopment Analysis to Spanish public universities and identified faculty quality and class availability as the binding constraints on timely graduation. [Gong et al. \(2024\)](#) surveyed 1,700 undergraduates across ten majors and showed that course-selection patterns and academic performance reinforce each other psychologically. [Yue and Fu \(2017\)](#) analyzed 12,000+ students over twelve years and concluded that academic performance and major-declaration timing dominate pre-college characteristics in predicting time-to-degree.

The literature converges on three points: workloads are unevenly distributed in many programs, students perceive that unevenness more accurately than credit hours capture it, and uneven distribution causally affects grades, well-being, and time-to-degree. None of the studies we are aware of, however, target a single engineering curriculum at the per-course level using direct study-time tracking, which is the gap this study addresses.

1.3 Research Questions and Hypothesis

This study addresses three questions: (1) Which courses in the ISE curriculum are objectively the most difficult, measured by the ratio of grade earned to time invested? (2) Is course difficulty distributed evenly across semesters, or are there “bottleneck” semesters that concentrate hard courses? (3) Do students agree with the case-study ranking, and would they prefer a different sequencing? The formal null hypothesis is that the Easiness Score is equal across all courses; the alternative is that at least one course’s Easiness Score differs significantly from at least one other.

2 Data

2.1 Data Sources and Participants

The study uses two data sources collected at the University of Oklahoma. The *case study* tracked one ISE student over five semesters from Spring 2022 through Spring 2024, recording daily time spent per course (in 0.1-hour increments via Pomofocus) and the final letter grade for each of 24 courses. After expanding daily logs across the active course-day grid, the case study yielded 2,668 observation rows. The *survey* was administered through Qualtrics to 24 upper-division (junior and senior) ISE students at OU; participants were recruited in person at common campus locations, and the instrument took roughly five minutes to complete. The full survey instrument is reproduced in Appendix F, and the 19 ISE-only courses presented as response options appear in Appendix C.

2.2 Variables and the Easiness Score

The primary outcome is the *Easiness Score*, defined per course as

$$ES_i = \frac{\text{Ending Grade}_i}{(\text{Hours studied per study day}_i) \times (\text{Number of study days}_i)}, \quad (1)$$

for course index i . Higher values indicate “easier” courses—i.e., higher grades earned per unit of study time. For the case study, the daily time logs make the denominator a random variable directly, so the Easiness Score itself is a random variable with a recoverable empirical distribution. For the survey, ending grades and study weeks are reported as scalar averages, while hours-per-study-week is reported as a categorical bin (Table 2); we sample 120 Monte Carlo replicates per course from the empirical hours-per-week distribution, divide by reported study weeks, and rescale by average ending grade to recover an analogous Easiness Score distribution.

2.3 Summary Statistics

Table 1 lists the 24 courses appearing in the case study with their acronyms used throughout. Tables 2 and 3 report the empirical hours-per-week distribution and the survey averages used to construct the survey-side Easiness Score for the five core ISE courses common to both phases.

Table 1: Case-Study Course Acronyms

Acronym	Course	Acronym	Course
AES	Applied Engineering Statistics	EiSD	Ergonomics in Systems Design
AR	Advanced R	FM	Fluid Mechanics
BDaH	Bhutan Democracy and Happiness	FoEE	Fund. of Engineering Economy
DDDM1	Data-Driven Decision Making I	GC	General Chemistry
DMP1	Design & Manufacturing I	GCL	General Chemistry Lab
DMP2	Design & Manufacturing II	LME	Leadership Mgt. Engineers
DSM	Deterministic Systems Models	MoM	Mechanics of Materials
EE	Enterprise Engineering	PD	Professional Development
EL	Experiential Leadership	PS/O	Production Systems / Ops
EfST	Entrepreneurship for Sci/Tech	PSM	Probabilistic Systems Models
Q&RE	Quality & Reliability Eng.	SAUS	Systems Analysis Using Sim.
SPDSS	Spreadsheet Decision Support	T	Thermodynamics

Twenty-four courses appear in the case-study log. The survey instrument restricted response options to the 19 ISE-listed courses (Appendix C).

The descriptive picture from Tables 2 and 3 is already informative. DDDM I has the lowest reported average ending grade (82) and a left-shifted study-hour distribution, while Q&RE shows the highest grade (91) on the shortest study commitment (6 weeks). The contrast suggests that grade and effort do not move together one-for-one, which is precisely why the Easiness Score—which integrates both—is the appropriate object of inference rather than either component alone.

Table 2: Empirical Distribution of Self-Reported Study Hours per Study Week (Survey)

Hours / Study Week	AES	DSM	DDDM1	Q&RE	EiSD
0.5	8%	0%	0%	0%	0%
1.5	17%	0%	4%	8%	25%
2.5	17%	0%	0%	17%	25%
3.5	21%	17%	17%	29%	8%
4.5	4%	17%	21%	21%	13%
5.5	21%	8%	17%	8%	13%
6.5	8%	17%	13%	17%	8%
7.5	0%	21%	17%	0%	4%
8.5	4%	21%	13%	0%	4%

Bin midpoints used for Monte Carlo sampling. Each course distribution sums to 100% across $n = 24$ survey respondents.

Table 3: Survey Inputs Used in Easiness-Score Calculations

Metric	AES	DSM	DDDM1	Q&RE	EiSD
Avg. Ending Grade	87	88	82	91	89
Number of Study Weeks	13.2	6.6	8.0	6.0	7.6

Averaged across $n = 24$ survey respondents. Combined with Table 2 via Equation (1).

2.4 Sample Construction

Case-study logs are complete by construction: the participant recorded a value for each tracked day, with zero entered for non-study days. The survey was distributed to a convenience sample of 24 ISE juniors and seniors recruited in person at common campus locations, and all 24 responses were retained. Both samples are voluntary and may over-sample students who self-select into a measurement exercise; we treat this as a limitation in Section 5.

3 Methodology

3.1 Empirical Framework

The case study is modeled as a single-factor within-subjects design with one participant and 24 treatment levels:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i \in \{1, \dots, 24\}, \quad j \in \{1, \dots, 112\}, \quad (2)$$

where Y_{ij} is the Easiness Score of course i on study-day j , μ is the grand mean, τ_i is the course effect, and ε_{ij} is unattributable error (which subsumes the participant effect since there is only one subject). The survey model is analogous but adds a participant effect:

$$Y_{ij} = \mu + \tau_i + \rho_j + \varepsilon_{ij}, \quad i \in \{1, \dots, 19\}, \quad j \in \{1, \dots, 24\}, \quad (3)$$

where ρ_j is the random effect for participant j and the 19 treatment levels are the ISE-only courses listed in Appendix C. In both models the null hypothesis is $\tau_1 = \tau_2 = \dots = \tau_K$, where $K = 24$ for the case study and $K = 19$ for the survey.

3.2 Statistical Tests

The Easiness Score is bounded below by zero and right-skewed (a few low-grade, low-effort courses produce extreme values), so we do not assume normality. The full procedure is:

1. **Shapiro-Wilk** for each course/semester separately, to confirm non-normality and justify a rank-based test.
2. **Kruskal-Wallis** as the omnibus test for H_0 . Appropriate for one factor with ≥ 3 treatment levels and within-subjects data.
3. **Bonferroni-adjusted Wilcoxon** pairwise tests, to identify which specific course pairs differ once the omnibus test rejects.

All tests are run in R; the case study has 24 courses ($\binom{24}{2} = 276$ pairwise comparisons), and the semester analysis has 4 semesters ($\binom{4}{2} = 6$ pairwise comparisons).

3.3 Monte Carlo Simulation

For the survey side, hours-per-week is reported categorically. To recover a distribution of Easiness Scores comparable to the case study, we draw 120 replicates per course from the empirical distribution in Table 2, multiply by the reported study-weeks scalar from Table 3, and divide the average ending grade by that product. The choice of 120 replicates provides sufficient mass to recover stable quantiles from the discrete bin structure without artificially smoothing the underlying distribution.

4 Results

4.1 Case Study: Course-Level Results

Figure 1 shows visible separation between the bottom and top of the ranking, motivating a formal test. The Shapiro-Wilk results in Table 4 confirm that 22 of 24 courses are non-normally distributed at the 5% level, supporting the rank-based approach.

The Kruskal-Wallis statistic is overwhelmingly significant ($p < 2.2 \times 10^{-16}$), so we reject the null of equal Easiness Scores. The post-hoc Bonferroni-Wilcoxon procedure flags 62 of 276 comparisons as significant; AES, DDDM1, and DSM appear in disproportionately many of these pairs as the “low” side, confirming visually what Figure 1 suggests.

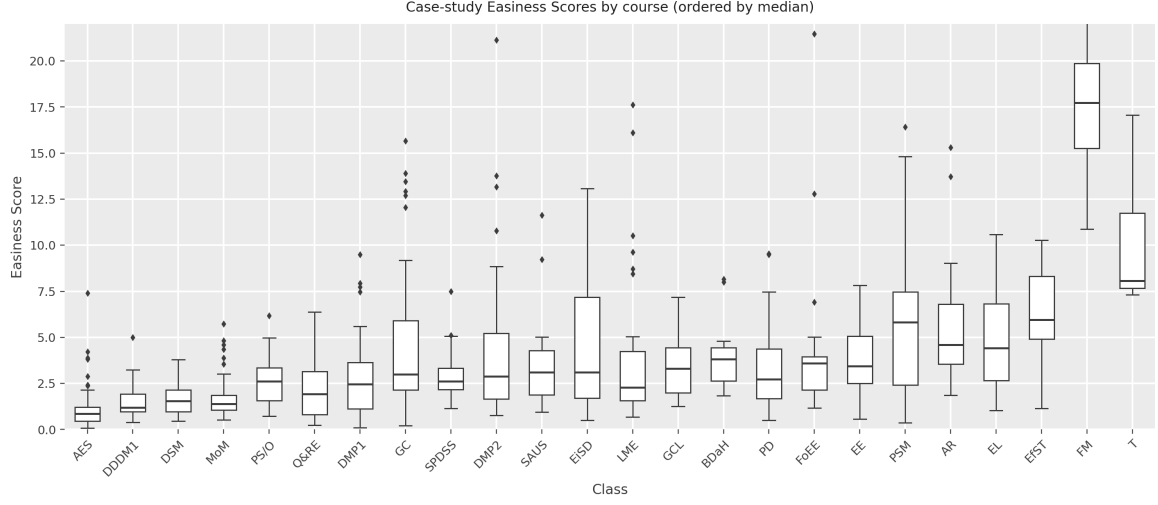


Figure 1: Boxplots of Easiness Scores across the 24 case-study courses, ordered by median. AES, DDDM1, DSM, MoM, and PS/O cluster at the bottom; T, FM, EfST, and EL sit at the top.

Table 4: Shapiro-Wilk Normality Test by Course (Case Study)

Course	p -value	Course	p -value	Course	p -value	Course	p -value
AES	6.91×10^{-9}	DSM	3.42×10^{-8}	FoEE	6.80×10^{-4}	PS/O	1.09×10^{-4}
AR	1.77×10^{-4}	EE	2.74×10^{-5}	GC	5.97×10^{-4}	PSM	2.56×10^{-7}
BDaH	5.97×10^{-4}	EL	8.06×10^{-4}	GCL	1.45×10^{-3}	Q&RE	2.98×10^{-6}
DDDM1	3.34×10^{-11}	EfST	1.45×10^{-2}	LME	4.27×10^{-5}	SAUS	2.95×10^{-6}
DMP1	5.61×10^{-7}	EiSD	2.55×10^{-8}	MoM	9.29×10^{-5}	SPDSS	1.10×10^{-4}
DMP2	1.51×10^{-6}	FM	3.84×10^{-1}	PD	5.09×10^{-3}	T	1.91×10^{-1}

Twenty-two of 24 courses reject normality at $\alpha = 0.05$. FM ($n = 7$) and T ($n = 8$) are the exceptions and have small samples.

Table 5: Kruskal-Wallis and Bonferroni-Wilcoxon Summary (Case Study, by Course)

Test	df	p -value	Pairwise Comp.	Significant
Kruskal-Wallis	23	$< 2.2 \times 10^{-16}$		
Bonferroni-Wilcoxon	—	—	276	62

The omnibus test rejects equality of Easiness Scores across courses. Sixty-two of 276 pairwise comparisons remain significant after Bonferroni correction at $\alpha = 0.05$; the full list appears in Appendix E.

4.2 Case Study: Semester-Level Results

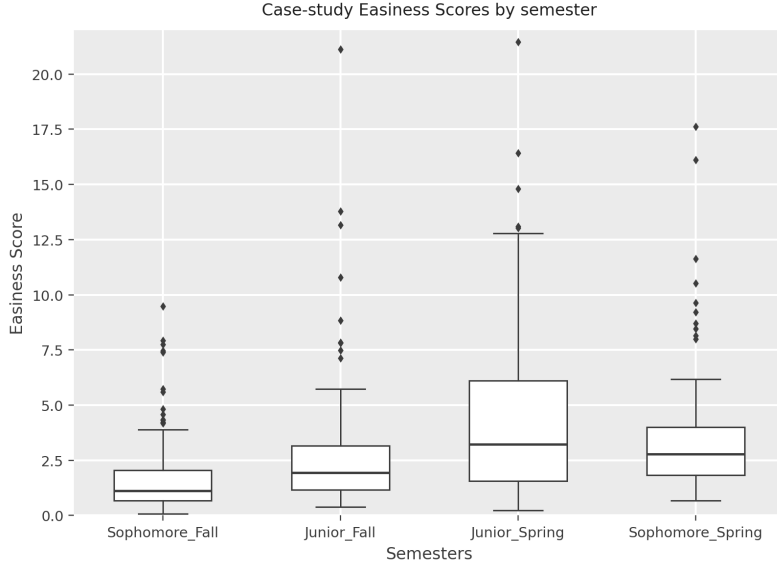


Figure 2: Easiness Scores aggregated by semester. Sophomore-Fall and Junior-Fall medians sit below those of the spring semesters, with Junior-Fall the most concentrated.

Table 6: Semester-Level Tests (Case Study)

<i>Panel A. Shapiro-Wilk</i>		<i>Panel B. Kruskal-Wallis</i>		
Semester	<i>p</i> -value	Test	df	<i>p</i> -value
Junior-Fall	4.96×10^{-13}	Kruskal-Wallis	3	5.6×10^{-5}
Junior-Spring	3.24×10^{-10}			
Sophomore-Fall	1.05×10^{-12}			
Sophomore-Spring	2.25×10^{-13}			

All four semester-level Easiness Score distributions reject normality. Kruskal-Wallis rejects equality of medians across semesters at $p = 5.6 \times 10^{-5}$.

Table 7: Bonferroni-Wilcoxon Pairwise Comparisons (Case Study, by Semester)

Treatment 1	Treatment 2	n_1	n_2	Test stat.	p_{adj}
Junior-Spring	Sophomore-Fall	76	89	4203	4.4×10^{-2}
Junior-Fall	Sophomore-Spring	82	84	2375.5	3.0×10^{-3}
Sophomore-Fall	Sophomore-Spring	89	84	2356	1.63×10^{-4}

Three of six pairwise semester comparisons remain significant after Bonferroni correction. The remaining three are not reported as they fail to reject at $\alpha = 0.05$.

The semester-level test confirms what the course-level results imply: Junior-Fall and Sophomore-Fall are statistically harder than Sophomore-Spring, with adjusted p -values below 0.005 in both cases. Of the four semesters tracked, Junior-Fall has the lowest median Easiness Score and the tightest interquartile range, consistent with the case-study student experiencing this semester as the bottleneck of the program.

4.3 Survey Results

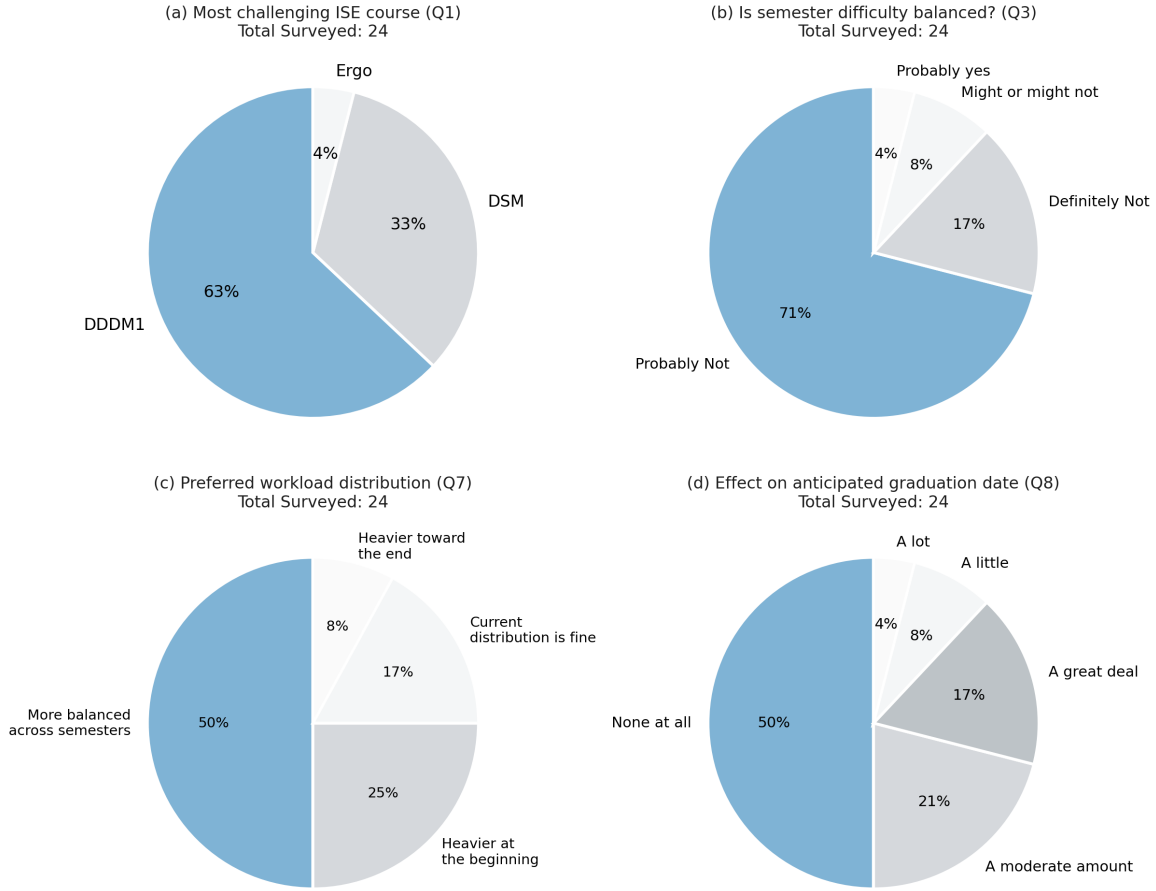


Figure 3: Survey responses on perceived difficulty and workload distribution. *Top*: most challenging ISE course (left), and whether semester difficulty is evenly balanced (right). *Bottom*: preferred workload distribution (left), and whether course distribution has affected anticipated graduation date (right). $n = 24$ for all panels.

The survey responses (Figure 3) reinforce the case-study findings. 63% of respondents identify DDDM I as the most challenging course in the curriculum, with DSM at 33% and Ergonomics at 4%. 71% answer “probably not” to whether semester difficulty is evenly balanced, with another 17% answering “definitely not” and only 4% answering “probably yes.” Half of respondents prefer a more even workload distribution across all semesters, and another 25% would prefer a heavier load at the beginning. Critically, 50% report that the course distribution has had at least some effect on their anticipated graduation date (with 17% answering “a great deal”).

Figure 4 disaggregates the perceived difficulty by semester. Sophomore-Fall and Sophomore-Spring concentrate at “Slightly” and “Moderately challenging,” Junior-Fall shifts decisively toward “Very” and “Extremely challenging” (9 respondents pick the latter), and Junior-Spring sits in between. This matches the case-study pattern in Figure 2 closely.



Figure 4: Distribution of self-reported difficulty by semester (5-point scale, $n = 24$ per semester). Junior-Fall is the only semester where “Extremely challenging” receives a substantial share of responses (9 of 24).

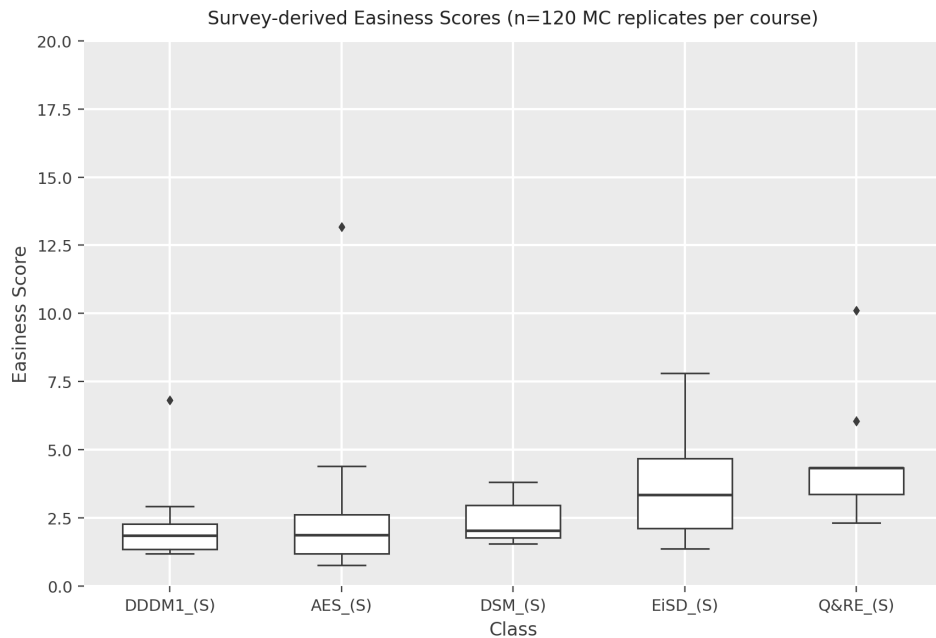


Figure 5: Boxplots of survey-derived Easiness Scores ($n = 120$ Monte Carlo replicates per course) for the five focal ISE courses. DDDM1, AES, and DSM cluster at the low end; EiSD and Q&RE at the top.

4.4 Cross-Validation: Case Study vs. Survey

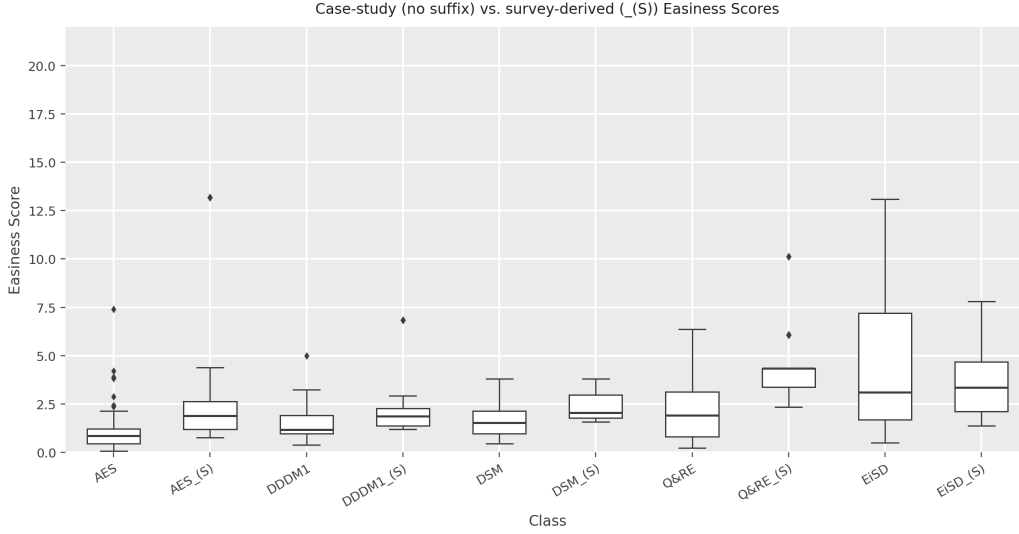


Figure 6: Side-by-side boxplots of Easiness Scores from the case study (no suffix) and survey ($_{(S)}$ suffix) for the five courses common to both phases. Median ranking is preserved across phases; survey distributions are tighter due to the $n = 120$ Monte Carlo replication.

Table 8: Case Study vs. Survey: Hardest-Courses Identification

Criterion	Case Study	Survey
3 hardest courses (by median)	AES, DDDM1, DSM	DDDM1, DSM, AES
≥ 1 course differs?	Yes ($p < 2.2 \times 10^{-16}$)	Yes
≥ 1 semester differs?	Yes ($p = 5.6 \times 10^{-5}$)	Yes
Hardest semester (standard track)	Junior-Fall	Junior-Fall
Notable observation	Spring semesters more dispersed	Ergonomics is an outlier

Both methods identify the same three hardest courses (with permutation in rank order) and the same hardest semester. The only meaningful divergence is dispersion structure across semesters.

The agreement across phases (Table 8) is the central robustness check of the study. Two completely different data-collection mechanisms—one tracking a single student in detail, one surveying 24 students at retrospective scale—identify the same three hardest courses and the same hardest semester. This convergence strongly suggests the findings are not artifacts of either methodology in isolation.

5 Conclusion

Across both the case study and the survey, Applied Engineering Statistics, Data-Driven Decision Making I, and Deterministic Systems Models emerge as the three hardest courses in the OU ISE curriculum, and Junior-year Fall is the hardest semester. The Kruskal-Wallis tests reject equality of Easiness Scores at $p < 2.2 \times 10^{-16}$ across courses and $p = 5.6 \times 10^{-5}$ across semesters, and 62 of 276 Bonferroni-adjusted pairwise comparisons remain

significant. 71% of surveyed students perceive workload as unevenly distributed, and 50% report that the distribution has affected their anticipated graduation date.

Recommendations. The clearest curriculum change suggested by these results is to separate DDDM I and DSM into different semesters, since both are concentrated in Junior-Fall and both rank among the three hardest courses. One feasible swap that respects existing prerequisites is to move DSM to Junior-Spring (where Probabilistic Systems Models currently sits) and shift PSM into the Junior-Fall slot DSM vacates. A secondary recommendation is to shift Applied Engineering Statistics and DDDM I back one semester each, so that students enter them with stronger calculus and probability foundations rather than concurrently. Together these adjustments remove the two heaviest courses from Junior-Fall as identified by the case-study Easiness Scores.

Limitations. The case study tracks a single subject, and the case-study student was forewarned by upperclassmen about specific courses (especially AES), which likely inflated study time relative to a counterfactual “naive” student. The survey relies on retrospective recall of study time committed sometimes years prior, which introduces measurement error in the denominator of the Easiness Score. Both phases use voluntary samples and may oversample students who self-select into a measurement exercise. The Easiness Score itself does not separate effort from grade; a low score could reflect either a hard course or a perfectionist student.

Future Research. The natural extension is to replicate the case study at scale by tracking a larger cohort of ISE students prospectively across the full junior-year sequence using the same daily time-logging protocol. Per-day tracking provides multiple observations per (subject, course) cell, which would let the participant random effect ρ_j in Equation (3) be identified separately from residual variance and disentangle course difficulty from student-level effort heterogeneity. A second extension is to construct an alternative degree flowchart that minimizes the variance of semester-level Easiness Scores subject to the prerequisite graph, framed as a constrained optimization problem.

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Appendix A. Case-Study Course Catalog

Table 9: Courses Tracked in the Case Study (Spring 2022 – Spring 2024)

Semester	Code	Course	Pattern
Sophomore-Spring '22	ISE-3293	Applied Engineering Statistics	MWF
	CEES-2153	Mechanics of Materials	MWF
	ISE-2303	Design & Manufacturing Process	TR
Junior-Fall '22	ISE-3304	Design & Manufacturing II	TR + lab
	ISE-4113	Spreadsheet Decision Support	TR
	ISE-4623	Deterministic Systems Models	TR
	ISE-2823	Enterprise Engineering	TR
	ISE-4553	Data-Driven Decision Making I	MW
Junior-Spring '23	ISE-4633	Probabilistic Systems Models	TR
	ISE-4223	Fund. of Engineering Economy	TR
	ISE-4563	Quality & Reliability Eng.	TR
	ISE-4804	Ergonomics in Systems Design	MWF
	ENGR-3051	Experiential Leadership	Async
Senior-Fall '23	ISE-4663	Systems Analysis Using Sim.	MW + lab
	ISE-4333	Production Systems / Ops	TR
	ENGR-4013	Leadership Mgt. Engineers	TR
	PSC-3953	Bhutan Democracy / Happiness	MW hybrid
Senior-Spring '24	CHEM-1315	General Chemistry (+ Lab)	MTWF + Mon lab
	ENGR-2002	Professional Development	MW
	ENGR-5122	Entrepreneurship Sci/Tech	T, half-term
	ENGR-2461	Thermodynamics	TR, half-term
	ENGR-3441	Fluid Mechanics	TR, half-term
	DSA-5041	Advanced R	Online, half-term

Full course descriptions and meeting locations available at classnav.ou.edu; ending grades and instructor information at judahs.site/ise.

Appendix B. Case-Study Time-Tracking Source

The raw daily time logs (one row per day, one column per active course) are stored in shared Google Sheets, one per semester:

- Sophomore-Spring 2022: [\[link\]](#)
- Junior-Fall 2022: [\[link\]](#)
- Junior-Spring 2023: [\[link\]](#)
- Senior-Fall 2023: [\[link\]](#)
- Misc. Spring 2024: [\[link\]](#)

Appendix C. ISE Courses Offered as Survey Response Options

Applied Engineering Statistics; Mechanics of Materials; Design and Manufacturing Processes; Design and Manufacturing II; Spreadsheet-Based Decision Support Systems; Deterministic Systems Models; Enterprise Engineering; Data-Driven Decision Making I; Probabilistic Systems Models; Engineering Economics; Quality and Reliability Engineering; Ergonomics in System Design; Systems Analysis Using Simulation; Production Systems & Operations; General Chemistry; General Chemistry Lab; Professional Development; Thermodynamics; Fluid Mechanics. (19 total.)

Appendix D. ISE Standard-Track Flowchart

The standard-track flowchart for the Bachelor of Science in Industrial & Systems Engineering (Standard Option B524, Summer 2020 – Spring 2021, 124 total credit hours) is published by the Gallogly College of Engineering and is available at [ou.edu](#) (PDF).

Appendix E. Significant Bonferroni-Wilcoxon Pairwise Comparisons (Case Study)

The 62 pairwise comparisons that remain significant after Bonferroni adjustment are listed below. Significance codes: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 10^{-4}$.

Treat. 1	Treat. 2	n_1	n_2	Stat.	p	p_{adj}	Sig.
AES	AR	66	12	83	1.49E-05	4.00E-03	**
AES	BDaH	66	14	128	2.38E-05	7.00E-03	**

Treat. 1	Treat. 2	n_1	n_2	Stat.	p	p_{adj}	Sig.
AES	DMP1	66	27	355	5.77E-06	2.00E-03	**
AES	DMP2	66	29	259	1.71E-08	4.72E-06	****
AES	EE	66	23	128	3.38E-09	9.33E-07	****
AES	EfST	66	9	9	2.72E-06	7.51E-04	***
AES	EiSD	66	38	365	1.96E-09	5.41E-07	****
AES	EL	66	16	36	8.71E-09	2.40E-06	****
AES	FM	66	7	1	1.69E-05	5.00E-03	**
AES	FoEE	66	22	156	4.01E-08	1.11E-05	****
AES	GC	66	38	587	6.75E-06	2.00E-03	**
AES	GCL	66	26	247	1.18E-07	3.26E-05	****
AES	LME	66	29	291	7.43E-08	2.05E-05	****
AES	MoM	66	39	684	6.40E-05	1.80E-02	*
AES	PD	66	27	161	6.52E-10	1.80E-07	****
AES	PS/O	66	34	540	2.30E-05	6.00E-03	**
AES	PSM	66	23	124	2.69E-09	7.42E-07	****
AES	Q&RE	66	30	369	9.26E-07	2.56E-04	***
AES	SAUS	66	21	140	4.17E-08	1.15E-05	****
AES	SPDSS	66	22	235.5	2.30E-06	6.35E-04	***
AES	T	66	8	2	5.23E-06	1.00E-03	**
DDDM1	DMP2	40	29	234	2.64E-05	7.00E-03	**
DDDM1	EE	40	23	139	4.68E-06	1.00E-03	**
DDDM1	EfST	40	9	19	3.35E-05	9.00E-03	**
DDDM1	EiSD	40	38	338.5	2.53E-05	7.00E-03	**
DDDM1	EL	40	16	47	7.57E-07	2.09E-04	***
DDDM1	FM	40	7	5	5.74E-05	1.60E-02	*
DDDM1	FoEE	40	22	148	1.76E-05	5.00E-03	**
DDDM1	PD	40	27	168	2.01E-06	5.55E-04	***
DDDM1	PSM	40	23	131	2.70E-06	7.45E-04	***
DDDM1	SAUS	40	21	145	3.03E-05	8.00E-03	**
DDDM1	T	40	8	4	1.66E-05	5.00E-03	**
DMP1	EfST	27	9	11	5.63E-05	1.60E-02	*
DMP1	EL	27	16	45	1.79E-05	5.00E-03	**
DMP1	FM	27	7	2	8.56E-05	2.40E-02	*
DMP1	T	27	8	3	3.87E-05	1.10E-02	*
DSM	EL	33	16	63	1.88E-05	5.00E-03	**
DSM	T	33	8	16	1.42E-04	3.90E-02	*

Treat. 1	Treat. 2	n_1	n_2	Stat.	p	p_{adj}	Sig.
EE	MoM	23	39	711	1.32E-04	3.60E-02	*
EE	PS/O	23	34	650	2.53E-05	7.00E-03	**
EfST	GC	9	38	324	3.56E-05	1.00E-02	**
EfST	MoM	9	39	334	2.89E-05	8.00E-03	**
EfST	PS/O	9	34	298	1.50E-05	4.00E-03	**
EiSD	T	38	8	19	1.18E-04	3.30E-02	*
EL	GC	16	38	540	7.85E-06	2.00E-03	**
EL	MoM	16	39	560	4.37E-06	1.00E-03	**
EL	PS/O	16	34	500	2.11E-06	5.82E-04	***
FM	GC	7	38	264	4.16E-05	1.10E-02	*
FM	GCL	7	26	177	1.60E-04	4.40E-02	*
FM	LME	7	29	196	1.66E-04	4.60E-02	*
FM	MoM	7	39	270	4.57E-05	1.30E-02	*
FM	PS/O	7	34	237	4.37E-05	1.20E-02	*
GC	PD	38	27	225	1.27E-04	3.50E-02	*
GC	T	38	8	4	1.81E-05	5.00E-03	**
GCL	T	26	8	10	1.41E-04	3.90E-02	*
LME	T	29	8	13	1.51E-04	4.20E-02	*
MoM	PD	39	27	214	4.63E-05	1.30E-02	*
MoM	T	39	8	6	2.22E-05	6.00E-03	**
PD	PS/O	27	34	745	3.29E-05	9.00E-03	**
PS/O	PSM	34	23	148	7.78E-05	2.10E-02	*
PS/O	T	34	8	2	1.77E-05	5.00E-03	**
SPDSS	T	22	8	8	1.80E-04	5.00E-02	*

Appendix F. Survey Instrument

1. Which ISE course do you believe to be the most challenging? (single-select: AES, DSM, DDDM I, Q&RE, EiSD, Other)
2. Rate each semester by level of difficulty. (5-point Likert: Not at all – Slightly – Moderately – Very – Extremely; one slider per semester for Sophomore-Fall, Sophomore-Spring, Junior-Fall, Junior-Spring)
3. Do you believe semester difficulty is evenly balanced across the ISE course plan? (5-point Likert: Definitely not – Probably not – Might or might not – Probably yes – Definitely yes)

4. In your most difficult ISE class, could you have received a higher grade if you had been able to commit more time to it? (5-point Likert: Strongly disagree – Strongly agree)
5. How much time did you spend on each focal course per week, in hours? (8-bin matrix per course)
6. What is your estimated ending grade in each focal course? (8-bin matrix per course)
7. How would you prefer the workload to be distributed across semesters? (single-select: More balanced; Heavier at the beginning; Heavier toward the end; Current distribution is fine)
8. How much has the distribution of course load in the ISE program influenced your anticipated graduation date? (5-point: None at all – A great deal)
9. How many weeks did you study for each focal course? (numeric input per course)

Live link: [Qualtrics survey](#).